Day 23: Binomial logistic regression

So far, every row of our data has been a *single* observation: one article, one student, one machine. Today each row is a **group** of observations that share the same predictor values, and the response is the **number** of successes in that group. This is **binomial logistic regression**.

Data: moths and natural selection

Researchers glued equal numbers of light and dark morph moths to tree trunks at 7 locations from 0 to 51.2 km from Liverpool, where the trees were blackened by pollution. After 24 hours, they recorded how many moths had been removed (presumably eaten by predators).

- `morph`: light or dark
- `distance`: kilometers from Liverpool
- `Placed`: number of moths of that morph placed at that location
- `Removed`: number of those moths removed after 24 hours

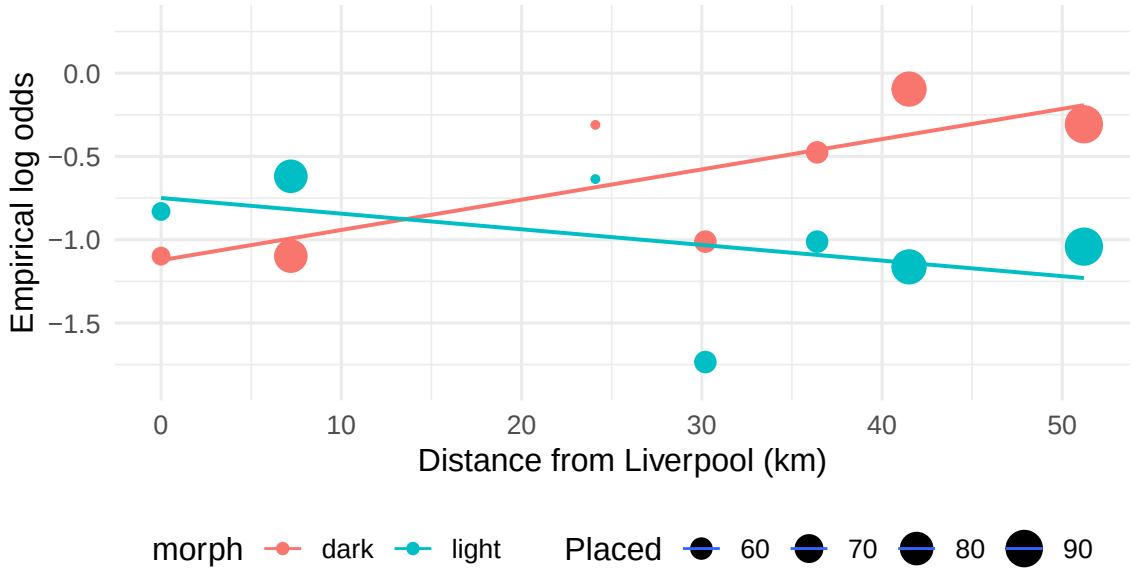
morph	distance	placed	removed	proportion removed
light	0.0	56	17	0.304
dark	0.0	56	14	0.250
light	7.2	80	28	0.350
dark	7.2	80	20	0.250
light	24.1	52	18	0.346
dark	24.1	52	22	0.423
light	30.2	60	9	0.150
dark	30.2	60	16	0.267
light	36.4	60	16	0.267
dark	36.4	60	23	0.383
light	41.5	84	20	0.238
dark	41.5	84	40	0.476
light	51.2	92	24	0.261
dark	51.2	92	39	0.424

Q1. What is the unit of observation (what does one row represent)? Why is this *not* the same kind of response as the fake news data?

Where we left off (HW8)

On the homework, we modeled the empirical log odds with `lm()`:

$$\text{log odds removed} = \log\left(\frac{\text{removed}}{\text{not removed}}\right) = \beta_0 + \beta_1\,\text{distance} + \beta_2\,\text{light} + \beta_3\,\text{distance} \times \text{light} + \varepsilon$$



Q2. What did we have to do by hand to make this work? Name two problems with this approach. (*Hint:* think about a row with 0 moths removed, and about the points on the plot above that are based on more moths than others.)

Why can't we just give `glm()` the count of moths removed?

```
glm(Removed ~ distance * morph, data = moths, family = "binomial")   # error
```

The response needs to be a 0/1 outcome or a **proportion** (with the number of trials supplied separately).

The binomial response

Let Y_i be the _____ of successes (here: moths removed) out of n_i trials (here: moths placed) in row i. We **assume** the n_i trials in each row are independent and have the same probability of success π_i.

Binomial distribution

The sum of n independent success/failure trials, each with success probability π, follows a **Binomial**(n, π) distribution:

$$E(Y) = _____ \qquad Var(Y) = _____$$

A binary response is the special case \$n = \$ _____.

Q3. The variance of a binomial count depends on π. What does that tell us about the "equal variance" condition we needed for OLS? Do we need to check it here?

The binomial logistic regression model

The model is the same as before: it describes how the **probability** of success depends on the predictors, using the logit link.

Log-odds scale:

$$\text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right) = _____ = \eta_i$$

Probability scale:

$$\pi_i = _____$$

Response distribution: $Y_i \sim$ Binomial(n_i, π_i) independently.

Instead of creating the log odds by hand, we give `glm()` the **proportion** removed as the response, and tell it how many trials each proportion is based on with `weights`:

```
model3 <- glm(Removed / Placed ~ distance * morph, weights = Placed,
              data = moths, family = "binomial")
```

(Equivalently, `cbind(Removed, Placed - Removed)` can be the response, with no `weights`.)

term	estimate	std. error	z	p-value
Intercept	-1.129	0.198	-5.70	<0.001
distance	0.019	0.006	3.28	0.001
morphlight	0.411	0.274	1.50	0.134
distance:morphlight	-0.028	0.008	-3.44	<0.001

Q4. Write the fitted model on the log-odds scale for **dark** moths and for **light** moths.

Dark:

Light:

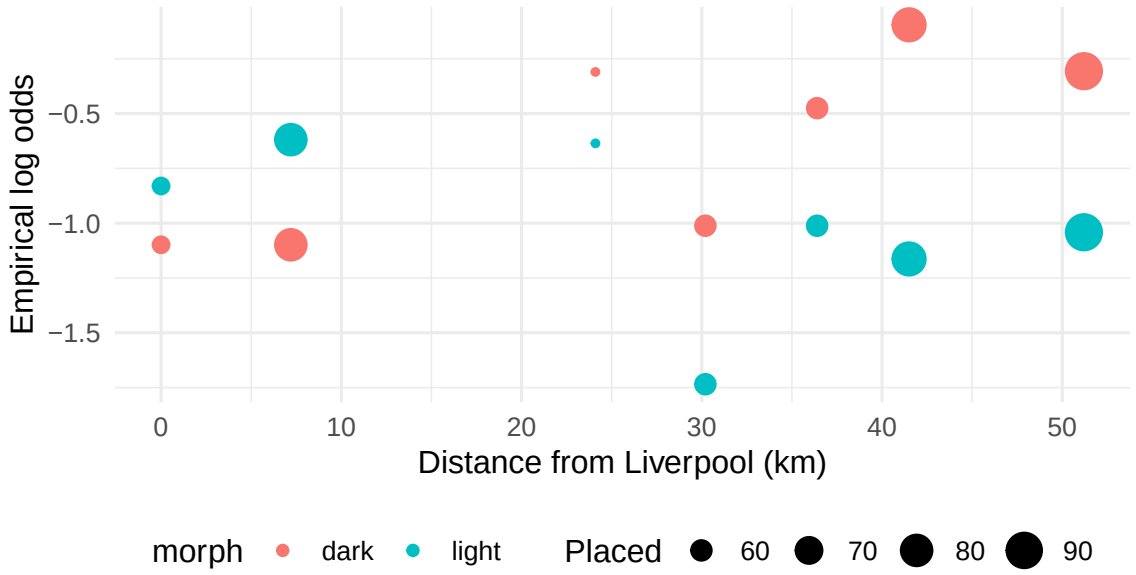
Q5. Interpret $e^{\hat{\beta}_1}$ (`distance`) in context. Which moths does this apply to?

Q6. For light moths, each additional km from Liverpool multiplies the odds of removal by _____. How does this compare to dark moths? What does this say about the hypothesis from the study?

Checking conditions

> **Binomial logistic regression conditions**
>
> - **Binomial counts**: Y_i is the number of successes in n_i trials
> - **Independence**: the trials (and the rows) are _____
> - **Linearity**: the _____ is a linear function of the predictors
> - **Variance structure**: $Var(Y_i) =$ _____, which is not constant, but is determined by the mean. We don't need an "equal variance" condition.

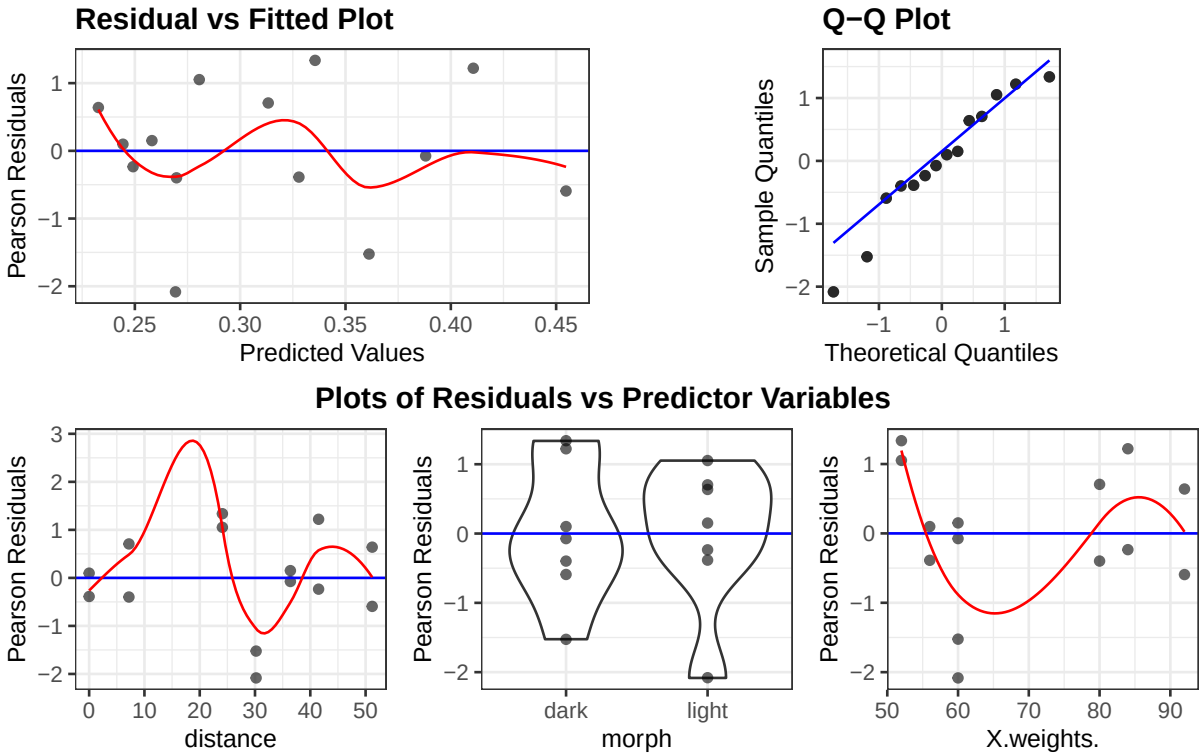
We check **linearity** with empirical logit plots (as in Day 20), but now every row is already a group, so we don't need to bin:



And we check linearity after accounting for the other predictors with **Pearson residuals**:

$$\text{Pres}_i = \frac{Y_i - n_i\hat{\pi}_i}{\sqrt{n_i\hat{\pi}_i(1-\hat{\pi}_i)}}$$

For large enough n_i, Pearson residuals behave approximately like draws from a _____ distribution (unlike binary logistic regression, where the residuals were in two bands). We plot them against the fitted values and the predictors, with a smoother.



Q7. What do you see in the Pearson residual plots? Is there evidence of nonlinearity or outliers? (The `X.weights.` panel is just the number of moths placed, which `glm()` stores as a predictor.)

Goodness of fit

Because each row is a group, we can ask a question we couldn't ask before: **does the model fit the group proportions?** The **deviance** compares the observed counts Y_i to the counts the model predicts, $n_i\hat{\pi}_i$:

$$D = 2\sum_i \left[Y_i \log\left(\frac{Y_i}{n_i\hat{\pi}_i}\right) + (n_i - Y_i)\log\left(\frac{n_i - Y_i}{n_i - n_i\hat{\pi}_i}\right)\right]$$

- D is _____ when the predicted counts are close to the observed counts (good), and _____ when they are far away (bad).
- If the model is correct and all the counts are large enough (about 5 or more successes *and* failures in every row), $D \overset{\cdot}{\sim} \chi^2$ with $df = n - (p+1)$, where n is the number of **rows** (binomial observations), not the number of moths.

Goodness-of-fit test

Hypotheses: H_0: _____ (the model is adequate) vs. H_a: the logit of π_i is *not* a linear function of the predictors (the model is inadequate)
Test statistic: D = the **residual deviance** reported by `summary()`
Reference distribution: χ^2 with $df =$ _____
p-value: the area _____ D (upper tail). A *small* p-value is evidence of lack of fit.

```
summary(model3)
```

```
Call:
glm(formula = Removed/Placed ~ distance * morph, family = "binomial",
    data = moths, weights = Placed)

Coefficients:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)         -1.128987   0.197906  -5.705 1.17e-08 ***
distance             0.018502   0.005645   3.277 0.001048 **
morphlight           0.411257   0.274490   1.498 0.134066
distance:morphlight -0.027789   0.008085  -3.437 0.000588 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

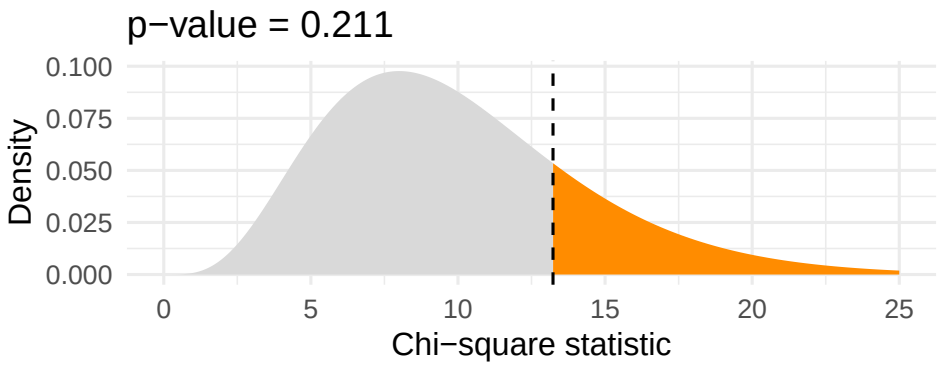
    Null deviance: 35.385  on 13  degrees of freedom
Residual deviance: 13.230  on 10  degrees of freedom
AIC: 83.904

Number of Fisher Scoring iterations: 4
```

Q8. How many binomial observations are there? How many parameters does the model have? Verify the residual degrees of freedom.

Q9. Calculate the p-value for the goodness-of-fit test, and state your conclusion in context.

```
1 - pchisq(____, ____)
```



> **Warning**
>
> This goodness-of-fit test is only valid for **binomial** logistic regression with large enough n_i. It is *not* valid for binary logistic regression (each row has $n_i = 1$).

Q10. The **likelihood ratio test** also uses deviances. What question does the LRT answer, and how is it different from the goodness-of-fit test? (*Hint:* what models are being compared?)

Q11. If the goodness-of-fit test had a very small p-value, what might be wrong with the model? List two things we could try.

Summary

	Binary logistic	Binomial logistic
Response		
`glm()` response		
Residual plots		
Goodness-of-fit test valid?		

- Binomial logistic regression models the _____ of success for each row, using the same logit link as binary logistic regression.
- The variance of Y_i is _____, so it depends on π_i.

- **Pearson residuals** behave like $N(0, 1)$ draws when the n_i are large, so we can plot them against fitted values and predictors.
- The **residual deviance** is compared to a χ^2 distribution with $n - (p + 1)$ df to test **goodness of fit**.

R function reference

Function	Purpose
`glm(y/n ~ x, weights = n, data, family = "binomial")`	Fit a binomial logistic regression (proportion response with `weights`)
`glm(cbind(y, n - y) ~ x, data, family = "binomial")`	Same model, with a (successes, failures) response
`resid_panel(model, type = "pearson", smoother = TRUE)`	Pearson residual plots
`resid_xpanel(model, type = "pearson", smoother = TRUE)`	Pearson residuals against each predictor
`summary(model)`	Residual deviance and df for the goodness-of-fit test
`1 - pchisq(D, df)`	Upper-tail χ^2 probability (p-value)